

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #2

Instructions

- TA in charge: Abidur Rahman (abidur.rahman@mail.mcgill.ca)
- Write/type your answers in the space provided.
- You must electronically submit the assignment in the form of a PDF file (only this format will be accepted) on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name:	
First student ID number:	
Second student full name:	
Second student ID number:	
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

Question 1 (12%)	Question 2 (05%)	Question 3 (13%)	Question 4 (10%)	Question 5 (10%)	Total
Question 6 (10%)	Question 7 (10%)	Question 8 (10%)	Question 9 (20%)		

Question 1 (12%)

The K-map of a 4-variable logic function $f(x_1, x_2, x_3, x_4)$ is

		x_2x_3			
		00	01	11	10
x_1x_4	00	0	0	1	0
	01	1	1	1	0
	11	0	1	1	1
	10	0	1	0	0

(a) List the prime implicants of the function f
(b) Group the prime implicants from (a) into essential and non-essential prime implicants
(c) Provide $\bar{f}$ (the complement of the function f) in SOP form

Question 2 (05%)

Provide the K-map of the following 5-variable logic function
 $g(x_1, x_2, x_3, x_4, x_5) = x_1'x_3'x_5 + x_1'x_3x_4 + x_1x_2'x_4x_5 + x_2x_4'x_5$

Question 3 (13%)

Consider the following Boolean function:

$$h(x_1, x_2, x_3, x_4) = \sum m(3, 7, 8, 9, 10, 11, 13, 15)$$

Find the simplest SOP of h , draw the circuit using only NAND and AND gates. Assume that the input variables are available in both true and complemented form at no additional cost.

Question 4 (10%)

Given the function $f = \sum (2, 4, 6, 8, 12) + \prod (1, 3, 13, 14)$

Consider the function $f = \sum m(2,4,6,9,12) + D(1,3,13,14)$

(a) Find the simplest SOP expression of the function using a K-map (show your work).

		$x_1 x_2$			
		00	01	11	10
$x_3 x_4$	00	0	1	1	0
	01	0	0	1	0
	11	0	0	0	0
	10	1	1	0	0

$f = x_2 \bar{x}_4 + \bar{x}_1 x_3 \bar{x}_4 + x_1 \bar{x}_3 x_4$

Cost = 15 = 4 gates + 11 inputs

(b) Find the simplest POS expression of the function using a K-map (show your work).

		$x_1 x_2$			
		00	01	11	10
$x_3 x_4$	00	0	1	1	0
	01	0	0	1	1
	11	0	0	0	0
	10	1	1	0	0

$f = (\bar{x}_2 + \bar{x}_4)(\bar{x}_1 + x_3)(x_2 + x_3 + x_4)$

Cost = 14 = 4 gates + 10 inputs

Question 5 (10%)

Consider the function $f = \sum m(1,3,4,9,11,13,15) + D(5,6,7,14)$

Find the simplest SOP and POS expressions of the function using K-maps (show your work).

SOP

		$x_1 x_2$			
		00	01	11	10
$x_3 x_4$	00	0	1	0	0
	01	1	0	1	1
	11	1	0	1	1
	10	0	0	0	0

$f = x_4 + \bar{x}_1 x_2$

Cost = 5 = 2 gates + 3 inputs

POS

		$x_1 x_2$			
		00	01	11	10
$x_3 x_4$	00	0	1	0	0
	01	1	0	1	1
	11	1	0	1	1
	10	0	0	0	0

$f = (\bar{x}_3 + x_4)(\bar{x}_1 + x_4)(x_1 + x_4)$

Cost = 13 = 4 gates + 9 inputs

Question 6 (10%)

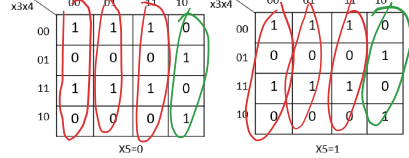
Given the K-map of a 5-variable function f below:

$x_1 x_2$		$x_3 x_4$	
00	01	10	11
0	1	0	1
1	0	1	0

A

$x_1 x_2$		$x_3 x_4$	
00	01	10	11
0	1	0	1
1	0	1	0

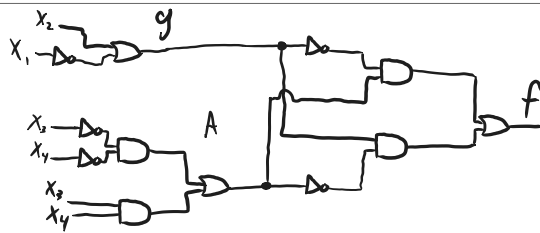
B



(a) Find the optimal (lowest cost) multi-level expression of f using functional decomposition.

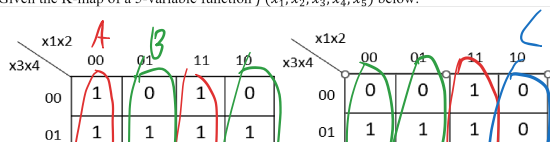
$$\begin{aligned}
 g &= (\bar{X}_1 + X_2) \rightarrow A & A &= (\bar{X}_3 \bar{X}_4 + X_3 X_4) \\
 g' &= X_1 \bar{X}_2 \rightarrow B & A' &= X_3 X_4 \text{ or } X_3 \bar{X}_4 \\
 & & A &= X_3 X_4 \text{ or } X_3 \bar{X}_4 \\
 & & B &= A' = (\bar{X}_3 \bar{X}_4 + X_3 \bar{X}_4) \\
 & & & \text{XOR} \\
 f &= g A + g' A' \\
 f &= g(\bar{X}_1 \bar{X}_2 + X_1 X_2) + g'(X_1 \bar{X}_2 + \bar{X}_1 X_2) \\
 \text{cost}(g) &= 4 \text{ (Not assuming line complement)} & B &= A' = (\bar{X}_3 \bar{X}_4 + X_3 \bar{X}_4) \\
 \text{cost}(A) &= 11 & & \\
 \text{cost}(f) &= \text{cost}(g) + \text{cost}(A) + 11 = 26 & &
 \end{aligned}$$

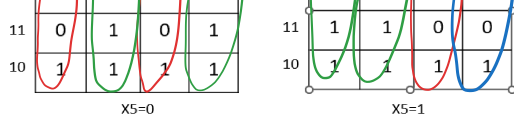
(b) Draw the circuit implementation from Q.1(a) with maximum fan-in equals to 2.



Question 7 (10%)

Given the K-map of a 5-variable function $f(x_1, x_2, x_3, x_4, x_5)$ below:





Using functional decomposition, find the expression of a multi-level implementation of f .
Hint: There could be more than two patterns in the K-map. Show your work and mark the patterns on the K-map.

$$\begin{aligned}
 g &= \bar{x}_5 \bar{x}_1 \bar{x}_2 + x_1 x_2 \rightarrow A = (\bar{x}_3 x_4) = \bar{x}_3 + x_4 \\
 h &= \bar{x}_1 x_2 + x_3 \bar{x}_1 \bar{x}_4 + \bar{x}_5 x_1 \bar{x}_2 \rightarrow B = x_3 + x_4 \\
 j &= x_5 x_1 \bar{x}_2 \rightarrow C = x_3 \bar{x}_4 \\
 f &= g(\bar{x}_3 + x_4) + h(x_3 + x_4) + j(x_3 \bar{x}_4)
 \end{aligned}$$

Question 8 (10%)

The VHDL code below describes the following system.

Inputs: 4-bit A and 2-bit D. Output: 1-bit Out

Functionality: When D = 00, Out = 0, When D = 01, Out = 1, if A is divisible by 2.

When D = 11, Out=1, if A is divisible by 3, When D = 10, Out=1, if A is divisible by 5. For all other cases, Out = 0.

Fill in the blanks of the provided code accordingly.

```

library IEEE;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity vhdliawesome is
    port (
        A : in std_logic_vector(3 downto 0);
        D : in std_logic_vector(1 downto 0);
        Out : out std_logic);
end vhdliawesome;

architecture description of vhdliawesome is
    signal div_by_2 : std_logic;
    signal div_by_3 : std_logic;
    signal div_by_5 : std_logic;
begin
    with A select div_by_2 <=
        '1' when "0000",
        '1' when "0010",
        '1' when "0100",
        '1' when "0110",
        '1' when "1000",
        '1' when "1010",
        '1' when "1100",
        '0' when OTHERS;

    with A select div_by_3 <=
        '1' when "0000",
        '1' when "0011",
        '1' when "0111",
        '1' when "1001",
        '1' when "1111",
        '0' when OTHERS;

    with A select div_by_5 <=
        '1' when "0000",
        '1' when "0101",
        '1' when "1010",
        '1' when "1111",
        '0' when OTHERS;

    with D select Out <=
        '0' when "00",
        div_by_2 when "01",
        div_by_3 when "11",
        div_by_5 when "10";
end description;

```

Question 9 (20%)

Use separate and joint optimization strategies to find out which one has the lowest-cost SOP implementation of f and g , draw the circuit and provide the cost in each case.

$$f(x_1, x_2, x_3, x_4) = \sum m(0,1,3,5,8,10,11) + D(7,14)$$

$$g(x_1, x_2, x_3, x_4) = \sum m(1,3,5,7,10,11,15) + D(0,9)$$

Note: The complements of the input variables (x_1, x_2, x_3, x_4) are given at no additional cost.

